

TEA FAZIO - ENGINEERING DESIGN PORTFOLIO

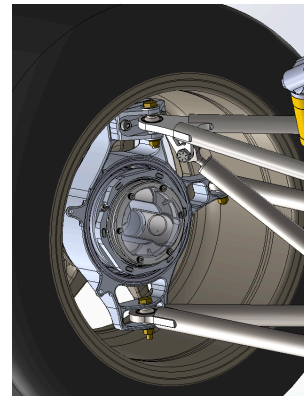
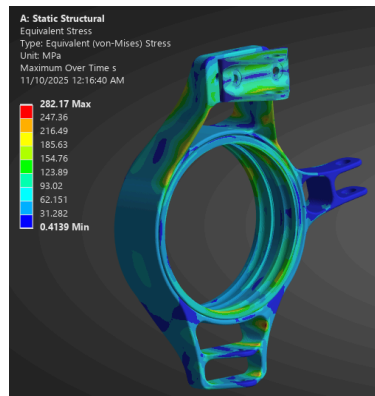
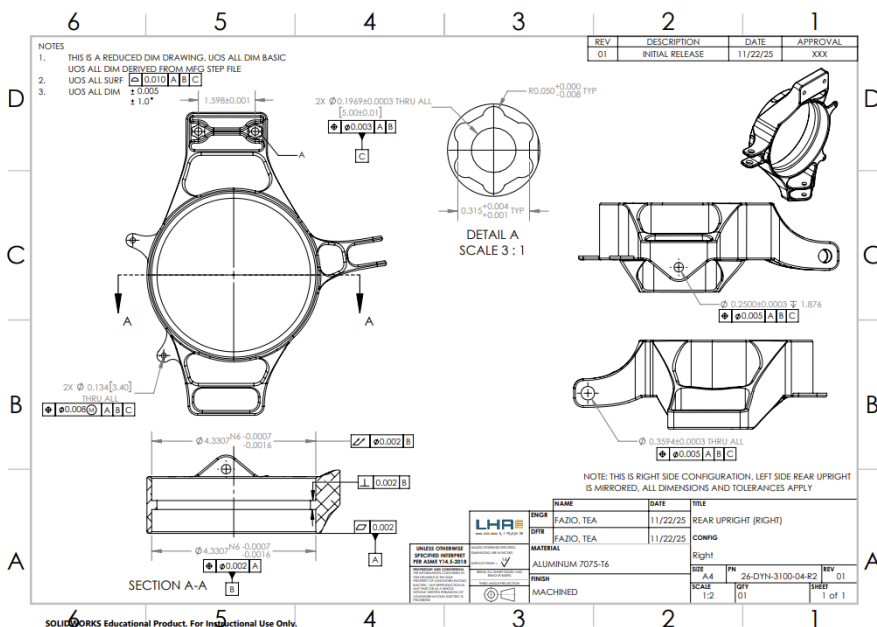
Mountain View, CA | teafazio@utexas.edu | www.linkedin.com/in/teafazio | (650) 450-1330



FSAE REAR WHEEL ASSEMBLY/ August 2025 - Present

Objective: Design a wheel upright component to interface with suspension linkages and wheel base, minimize bearing compliance and weight while maximizing stiffness.

- Specped bearings based on target axial and radial loads from wheel contact patch.
- Maintained same weight as 2024 vehicle design while improving wheel bearing stiffness by 15%
- Iterated design through ANSYS FEA software to minimize stress concentrations and ensure all designs provided 2x factor of safety under a 2-2-3g load case.
- Analyzed wheel hub fatigue under typical driving load cases and additional vibrations from vehicle transport.

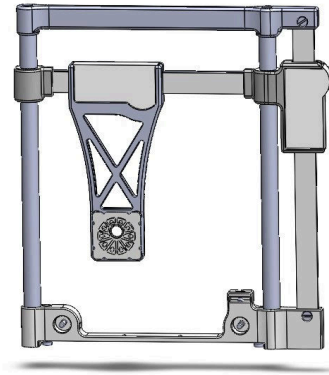
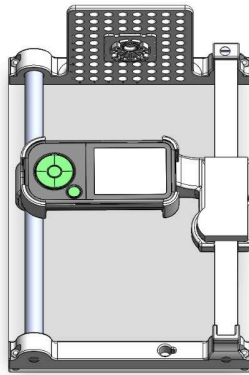
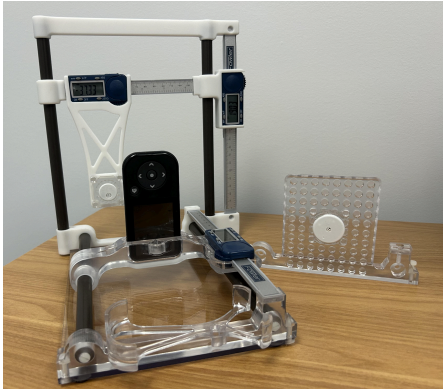


Top Skills: Design for Manufacturing, ANSYS FEA, Solidworks, GD&T

NFC TEST FIXTURE / May 2025 - August 2025

Objective: Design a device to test NFC readability range in fixed positions and continuous range on all 6 sides of remote, minimizing interference and with a 0.5 mm precision range.

- Designed a precision Z-axis sliding mechanism using PTFE bushings and Delrin rods to enable smooth, continuous motion.
- Interchangeable back plate system allows switching between a fixed grid configuration and a 3-axis gantry.
- Remote secured in front/back slot positions and clips with indexed pin/slot alignment to detect readings from side positions.
- Modified digital calipers to provide accurate linear measurements with adjustable zero-point calibration.
- Manufactured all components with controlled tolerance stackups through external suppliers to ensure consistent mechanical fit and repeatability.
- Validated system performance through prototyping to achieve measurement accuracy within ± 0.1 mm specifications.

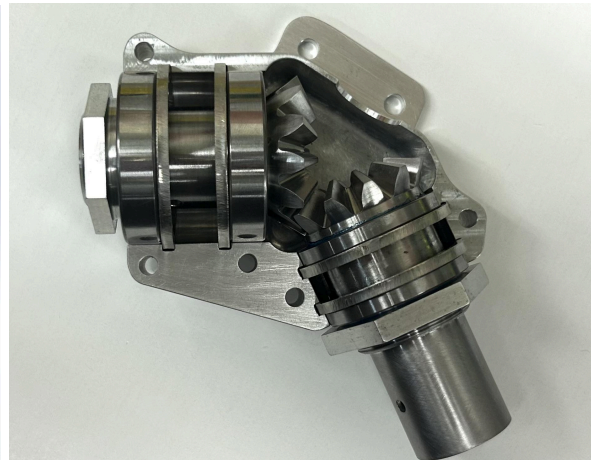
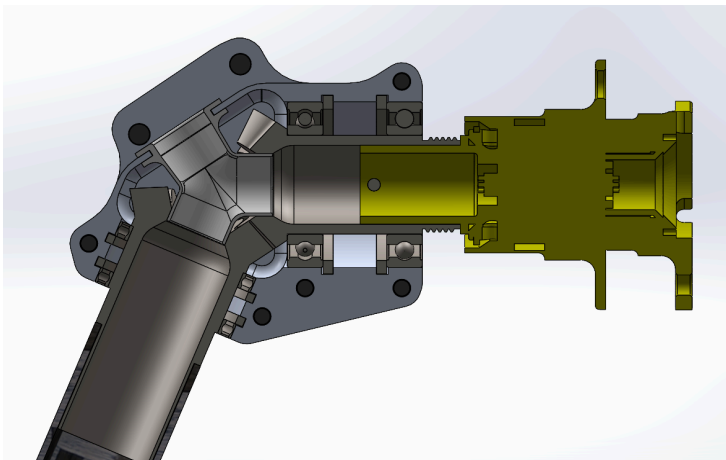


Top Skills: 3D printing, Electrical Integration, Solidworks, Design for Assembly

STEERING GEARBOX / September 2024 - May 2025

Objective: Design and a steering system with minimal (< 5 deg) backlash and stiffness for an FSAE Electric vehicle.

- Designed split housing assembly containing custom bevel gears, preloaded with threaded adjustment nut and mounted to carbon fiber steering column.
- Analyzed loading conditions to withstand maximum driver input forces and torques; performed manual calculations to select bearings that achieved 15% weight reduction from previous assembly while maintaining equivalent stiffness.
- Conducted torque testing on carbon fiber steering column to measure deflection under peak steering loads with 4x FOS.
- Created 3D models in SOLIDWORKS CAD software and validated performance using ANSYS finite element analysis.
- Produced components using in-house CNC lathe, CNC mill, and manual machining; coordinated external 5-axis CNC manufacturing with detailed GD&T drawings.

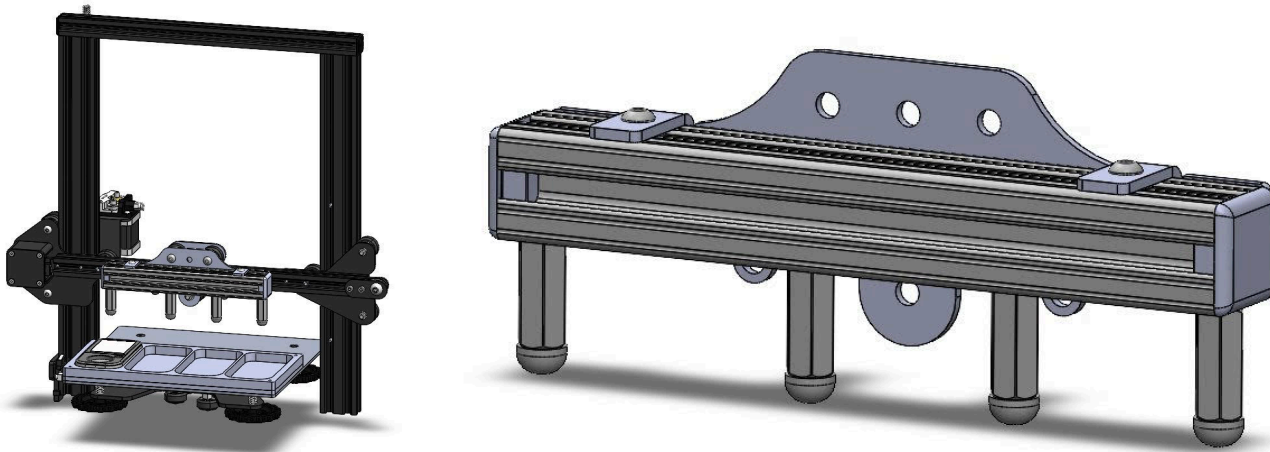


Top Skills: Solidworks, Ansys, GD&T, CNC Lathe, Manual Mill, Design for Manufacturability, Decision Matrices

BUTTON LIFE CYCLE TESTER / May 2025 - August 2025

Objective: Create a device to press all 6 buttons on a remote upwards of 100,000 times to test its life cycle

- Designed a robust sheet metal fixture optimized for long-term stability and minimal wear over 100,000+ automated test cycles.
- Modified an Ender 3 3D printer by removing the extruder assembly and repurposing the XY carriage to actuate button presses across six programmed positions.
- Developed a custom Python script to convert CSV input data into G-code commands, automating test cycles and coordinate mapping for each button location.
- Reduced cycle time by 75% through simultaneous testing of four devices, creating a more scalable and automated solution than previous manual methods.

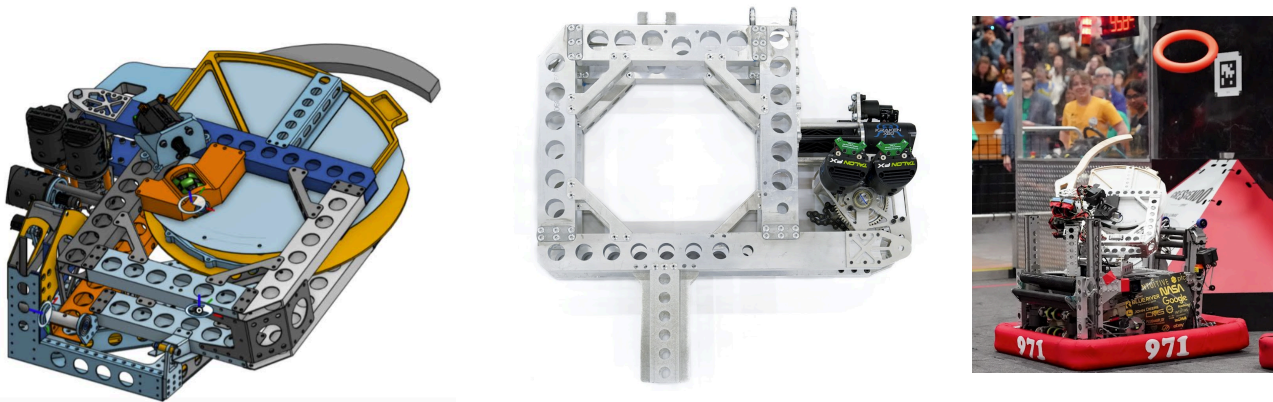


Top Skills: Python, Sheet metal, 3D Printing, Solidworks, Design for Durability

CATAPULT SUBSYSTEM / January - April 2024

Objective: Design a system to launch a foam ring at high speeds into a ~6-inch gap from varying distances.

- Designed 3 integrated custom gearbox systems for the assembly:
 - Pivoting mechanism utilizing 3D-printed herringbone gears for smooth angular motion.
 - High-speed catapult actuator employing multiple motors for rapid arm deployment.
 - Retaining wheel system to secure foam rings during dynamic movement and impact events.
- Utilized ONSHAPE sheet metal features and generated ANSI-compliant technical drawings for external manufacturing.
- Machined all gearbox shafts in-house using CNC lathe operations to ensure precision and quality control.
- Developed protective enclosures for potentiometer, beam break sensor, and magnetic encoder components.
- Prototyped approximately 15 arm configurations and tested 4 plate materials to optimize rotation while minimizing friction.



Top Skills: Onshape, CNC lathe, AANSI Drawings, Sheet Metal, Prototyping, Electrical Enclosures.

DRIVER STATION SUBSYSTEM / January - April 2023

Objective: Design an intuitive driver interface for efficient robot operation.

- Optimized button layout through operator consultation to improve ergonomics and workflow efficiency.
- Designed control panel in Onshape CAD and manufactured components using CNC router machining.
- Collaborated with electrical teams to integrate buttons, joysticks, and potentiometers into unified control system.
- Developed modular interface architecture enabling rapid reconfiguration for annual design iterations.



Top Skills: Onshape, Electrical Integration

2023 FRC 971 ROBOT / January - April 2023

Objective: Managed the development of a 5-DOF robotic arm with 360° vision, completed in under 6 weeks.

- Acted as project manager, delegating tasks to software, mechanical, and electrical teams.
- Oversaw the design, manufacturing, and assembly of the entire robot ahead of schedule.
- Led testing during team practices and was able to quickly respond and repair issues.

Top Skills: Project Management, Product Management, Communication, Failure analysis, Logistics

